CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES

41. Which of the following elements show the given properties?
(i) All elements are metals.
(ii) Most of the elements form coloured ions, exhibit variable valence and paramagnetism.
(iii) Frequently used as catalysts.

NCERT Page 84 / N-84

(a) Chalcogens
(b) Transition elements
(c) Inner transition elements
(d) Representative elements

42. Element X forms a chloride with the formula XCl_2, which is a solid with a high melting point. X would most likely be in the same group of the periodic table as

NCERT Page 85 / N-85

(a) Na
(b) Ca
(c) Al
(d) Si

43. Which is a metalloid?

NCERT Page 85 / N-85

(a) Pb
(b) Sb
(c) Bi
(d) Zn

44. Which of the following elements are found in pitch blende?

NCERT Page 78 / N-78

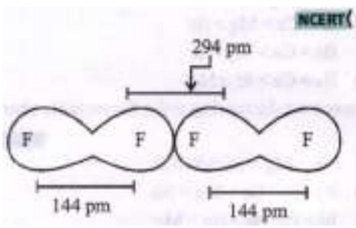
(a) Actinium and protoactinium
(b) Neptunium and plutonium
(c) Actinium only
(d) Both (a) and (b)

45. The order of increasing sizes of atomic radii among the elements O, S, Se and As is

NCERT Pages 85–86 / N-85–86

(a) As < S < O < Se
(b) Se < S < As < O
(c) O < S < As < Se
(d) O < S < Se < As

46. The van der Waals and covalent radii of fluorine atom respectively, from the following figure, are



(a) 219 pm, 72 pm
(b) 75 pm, 72 pm

(c) 147 pm, 72 pm

(d) 147 pm, 144 pm

47. As we move across the second period from C to F, ionisation enthalpy increases, but the trend from C to F for ionisation enthalpy is C < O < N < F. This is because

NCERT Page 88 / N-88

(a) the atomic radius of O is greater than the atomic radius of N

(b) the electronic configuration of N is more stable than the electronic configuration of O

(c) the atomic radius of N is greater than the atomic radius of O

(d) none of these

48. Covalent radii of atoms vary in the range of 72 pm to 133 pm from F to I, while those of noble gases from He to Xe vary from 120 pm to 220 pm. This is because, in the case of noble gases

NCERT Page 86 / N-86

(a) covalent radius is very large

(b) van der Waals radius is considered

(c) metallic radius is considered

(d) none of these

49. Why is the size of an anion larger than that of the parent atom?

NCERT Page 87 / N-87

(a) Due to increased repulsion among the electrons.

(b) Due to decrease in effective nuclear charge.

(c) Due to increase in effective nuclear charge.

(d) Both (a) and (b)

50. In the ions P^{3-}, S^{2-} and Cl^{-}, the increasing order of size is

NCERT Page 87 / N-87

(a) $Cl^{-} < S^{2-} < P^{3-}$

(b) $P^{3-} < S^{2-} < Cl^{-}$

(c) $S^{2-} < Cl^{-} < P^{3-}$

(d) $S^{2-} < P^{3-} < Cl^{-}$

51. Arrange the following in increasing order of ionic radii: C^{4-}, N^{3-}, O^{2-} and F^{-}.

NCERT Page 87 / N-87

(a) $C^{4-} < N^{3-} < O^{2-} < F^{-}$

(b) $N^{3-} < C^{4-} < O^{2-} < F^{-}$

(c) $F^{-} < O^{2-} < N^{3-} < C^{4-}$

(d) $O^{2-} < F^{-} < N^{3-} < C^{4-}$

52. The correct order of melting point is

(a) Be > Mg > Ca > Sr

(b) Sr > Ca > Mg > Be

(c) Be > Ca > Mg > Sr

(d) Be > Ca > Sr > Mg

53. The correct decreasing order for metallic character is

NCERT Page 91 / N-91

(a) Na > Mg > Be > Si > P

(b) P > Si > Be > Mg > Na

(c) Si > P > Be > Na > Mg

(d) Be > Na > Mg > Si > P

54. Arrange S, P and As in order of increasing ionisation energy.

(a) S < P < As

(b) P < S < As

(c) As < S < P

(d) As < P < S

55. Of the given electronic configurations for the elements, which electronic configuration indicates that there will be an abnormally high difference in the second and third ionisation energies for the element?

NCERT Page 88 / N-88

(a) $1s^2\ 2s^2\ 2p^6\ 3s^2$

(b) $1s^2\ 2s^2\ 2p^6\ 3s^1$

(c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

(d) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$

56. Which of the following statement(s) is/are incorrect?

(i) Ionisation enthalpy is expressed in units of kJ mol^{-1}.

(ii) Ionisation enthalpy is always positive.

(iii) Second ionisation enthalpy will be higher than the third ionisation enthalpy.

NCERT Page 88 / N-88

(a) Only (ii)

(b) Only (iii)

(c) (ii) and (iii)

(d) None of these

57. The first ($\Delta_i H_1$) and second ($\Delta_i H_2$) ionisation enthalpies (in kJ mol^{-1}) and the electron gain enthalpy ($\Delta_{eg}H$) (in kJ mol^{-1}) of the elements I, II, III, IV and V are given below.

NCERT Page 88 / N-88

Element	$\Delta_i H_1$	$\Delta_i H_2$	$\Delta_{eg}H$
I	520	7300	−60
II	419	3051	−48
III	1681	3374	−328
IV	1008	1846	−295
V	2372	5251	+48

The most reactive metal and the least reactive non-metal of these are respectively

(a) I and V

(b) V and II

(c) II and V

(d) IV and V

58. Among the following transition elements, pick out the element/elements with the highest second ionisation energy.

(A) V (At. no. 23)
(B) Cr (At. no. 24)
(C) Mn (At. no. 25)
(D) Cu (At. no. 29)
(E) Zn (At. no. 30)

NCERT Page 88 / N-88

(a) (A) and (C)

(b) (B) and (D)

(c) B and E

(d) Only (D)

59. Halogens and chalcogens have highly electron-gain enthalpy values that are generally

NCERT Pages 89–90 / N-89–90

(a) negative

(b) positive

(c) zero

(d) infinity

60. The correct order of electron gain enthalpies of Cl, F, Te and Po is

NCERT Pages 89–90 / N-89–90

(a) F < Cl < Te < Po

(b) Po < Te < F < Cl

(c) Te < Po < Cl < F

(d) Cl < F < Te < Po